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Generalization Across Multiple Mathematical Areas

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ABSTRACT

The recommendation to make generalization a central component of mathematics instruction from elementary school through undergraduate mathematics poses serious challenges in light of the research base that identifies students' difficulties in creating and expressing correct mathematical generalizations and the challenges teachers face in supporting students' abilities to generalize. Furthermore, although student difficulties are well documented, the instructional conditions necessary for fostering generalization are not well understood, particularly at the secondary and undergraduate levels. This project addresses these challenges by developing a comprehensive framework characterizing productive mathematical generalization in Grades 8-16 and identifying instructional interventions that can support correct generalizing. The project occurs within multiple mathematical domains extending from middle school to undergraduate mathematics, including algebra, geometry, calculus, and combinatorics. The project investigators will leverage student interviews, teaching experiments, and design experiment methodologies in order to characterize the processes of generalizing and to identify the instructional conditions that support productive generalization. The results of the project will identify

specific tasks and activities fostering student generalizing in a diversity of mathematical settings, which will be of practical use to teachers, school districts, teacher educators, and university instructors.

Mathematical generalization, the ability to create general rules, formulas, and strategies, is a key aspect of doing mathematics. Policy makers recommend making generalization a central component of mathematics instruction at every grade level from elementary school through undergraduate mathematics, with the Common Core State Standards highlighting generalization as a major goal in both the content and the practice standards. However, these recommendations pose serious challenges given students' pervasive difficulties in creating and expressing generalizations. In a report on performance assessments from more than 60,000 secondary students, findings revealed only a 20% success rate in students' creation of correct general statements. Students' challenges with mathematical generalizations also contribute to difficulties in mathematics achievement in many domains, including algebra, geometry, and combinatorics. This project will address these challenges by investigating how students generalize productively and how teachers can support more effective mathematical generalization. Through student interviews and teaching experiments, the project investigators will explore these issues in algebra, geometry, calculus, and combinatorics. Student participants will range from middle school students through undergraduates. The diverse range of student ages and mathematical domains will contribute to a robust model characterizing how students generalize in Grades 8-16. These findings will also identify instructional activities that can better support generalization in many different settings, which will be of use to teachers, school districts, teacher educators, and university instructors. The knowledge generated from the project will support improved student performance in critical areas of undergraduate mathematics, thus contributing to a diverse and globally competitive STEM workforce.

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PROJECT OUTCOMES REPORT

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The outcomes from the project entitled *Generalization Across Multiple Mathematical Areas* offer four novel contributions to the fields of mathematics education, teacher education, and the learning sciences: (1) An improvement of the field's understanding about the nature of students' mathematical generalizations across multiple domains and grade bands; (2) An identification of a set of domain-specific findings for supporting students' generalizations about algebraic functions and relations; (3) An identification of findings for fostering students' generalizations about discrete mathematics and combinatorics; and (4) The development of a set of general design principles characterizing the task features and instructional moves that are effective in fostering productive mathematical generalization.

The first outcome addresses the cognitive processes that can support productive mathematical generalizing. Generalization, or the development of a general principle, rule, or strategy, is a critical component of mathematical reasoning, with researchers recommending that it be central to education at all grade levels. However, research on students' generalizing reveals pervasive difficulties in creating and expressing correct general statements, which underscores the need to better understand the processes that can support more productive generalizations. In response, we conducted 146 interviews with 93 participants in middle-school through college in the domains of algebra, advanced algebra, trigonometry/pre- calculus, and combinatorics while solving complex problems. Our findings yielded the Relating- Forming-Extending (RFE) Framework, which distinguishes multiple related forms and types of generalizing across grade bands and content domains. The resulting structure provides a way to understand how students can shift between forms, as well as engage in multiple types of generalizing within each form, when immersed in the ongoing process of developing and refining generalizations over time. The RFE Framework extends the research base on generalizing into more advanced mathematical domains, as well as incorporating more open-ended and complex tasks to afford an exploration of how students generalize when engaged in more naturalistic problem-solving contexts.

The second and third outcomes address the types of mathematical activity that support productive generalizing in algebraic reasoning and combinatorial reasoning. We examined the findings from 13 design experiments with a total of 52 middle-school, high-school, and undergraduate students in order to identify effective mathematical interventions in each of these domains. In the areas of algebra, advanced algebra, and trigonometry/precalculus, we found that that providing students with opportunities to reason with co-varying quantities meaningfully supported their abilities to make powerful generalizations about functional relationships. Students explored linear, quadratic, higher-order polynomial, and trigonometric relationships by investigating the ways in which two quantities simultaneously varied together in a dependency relationship. Quantities are people's conceptions of measurable attributes of objects or events, such as distance, length, time, volume, or temperature. When exploring how quantities changed in tandem, students were able to develop and generalize more robust functional relationships, particularly in relation to average and instantaneous rates of change, which are key ideas in calculus and beyond.

In the areas of combinatorial reasoning and discrete mathematics, we found that students engaged in productive generalizing when they were given opportunities to reflect on and extend their initial combinatorial activity. Students naturally connect back to prior work they have done, leveraging relating activity to solve new problems and justify new problem-solving approaches. This highlights the importance of having students engage in rich initial combinatorial thinking (such as listing, focusing on sets of outcomes, or justifying formulas combinatorially), and it suggests that fostering meaningful initial activity can facilitate useful and effective generalizing. We also developed effective combinatorial tasks that increased in complexity and generality. We found that students could complete such tasks successfully, particularly when they developed combinatorial meanings and justifications for each part of the task along the way. Finally, we investigated students' work with combinatorial proof, and we found that an effective intervention was to have students first try to solve specific, concrete problems, with specific numbers and contexts, which they could then extend to give proofs of generally-stated binomial identities.

Our final outcome addresses the instructional moves and task design features that have been shown to be the most effective in fostering productive mathematical generalizations. We have identified six major design principles for developing and implementing instruction: a) Prompt reflection on work to articulate and present similarities and differences in students' actions; b) Design tasks that will perturb prior problematic generalizations; c) Require prediction without allowing calculation; d) Implement situations with surface similarities that contain structural differences; e) Create repeated cycles of relating, forming, and extending within and across tasks; and f) Elicit domain-specific activity for generalizing. These findings offer a series of practical suggestions that teachers and university instructors can implement to more effectively support their students' generalizing activity.

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